SURFACE BEHAVIOUR OF SOIL COLLOIDS

Course: Soil Colloids and Minerals

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Introduction

Soil is a complex natural system made up of mineral particles, organic matter, water, air and living organisms. The mineral particles are commonly divided into sand, silt and clay according to size. Among these components, clay and humus are especially important because they behave as colloids and provide most of the chemically active surface in soil.

The surface of a soil colloid is the boundary between a solid particle and the surrounding soil solution. This boundary is not inactive. Ions may be adsorbed, exchanged or released; water molecules may become oriented around charged sites; and organic compounds may form strong associations with minerals. Many reactions that determine soil fertility and soil structure occur at this solid–solution interface.

Soil colloids are very small, and their surface area per unit mass is therefore very high. A large surface area provides many sites for the retention of nutrient ions, water, microorganisms and pollutants. Most clay and humus colloids have a net negative charge and attract cations such as calcium, magnesium, potassium, ammonium, hydrogen and aluminium. Some iron and aluminium oxides, allophane and mineral edges can become positively charged under acidic conditions and attract anions such as phosphate and sulphate.

Surface behaviour refers to the physical and chemical processes that occur at or near colloidal surfaces. The major processes are adsorption, desorption, ion exchange, hydration, formation of the electrical double layer, flocculation, dispersion, swelling and shrinkage. These processes are affected by mineral type, particle size, pH, electrolyte concentration, exchangeable ions, organic matter and moisture content.

The study of surface behaviour is important for agriculture, environmental science and engineering. It explains why some soils retain nutrients while others lose them by leaching, why some clays form stable aggregates while others disperse, why phosphorus may be strongly fixed in acidic soils, and why smectite-rich soils expand when wet and contract when dry. The purpose of this assignment is to explain these surface processes and connect them with practical soil management. (Iowa State University Digital Press, 2025; Moorberg & Crouse, 2021)

Abstract

Abstract. Soil colloids are the most reactive part of soil. They include crystalline silicate clays, poorly crystalline minerals, iron and aluminium oxides, and organic colloids. Their very small size creates a large specific surface area. Their electrical charge allows them to attract ions and water and to participate in adsorption and exchange reactions. As a result, colloids control nutrient retention, pH buffering, water storage, aggregate formation, dispersion, swelling and shrinkage. This assignment discusses the origin of colloidal surface charge, the electrical double layer, adsorption, ion exchange, cation exchange capacity, anion retention, flocculation and dispersion. It also explains the importance of colloidal behaviour in soil fertility, irrigation, erosion control, pollution management and construction. (Iowa State University Digital Press, 2025; Moorberg & Crouse, 2021)

The central idea is that the behaviour of a soil depends strongly on the properties of its solid–solution interface. A change in pH, salt concentration or exchangeable cation can change the charge and interaction of colloidal particles. Understanding these processes is therefore essential for maintaining productive and stable soil.

1. General Nature of Soil Colloids

A colloid is a very small particle dispersed in a medium. Soil colloids are mainly associated with the fine clay fraction and with humus. Clay is commonly defined as material smaller than 2 micrometres, although colloidal behaviour depends on surface activity as well as size. Clay-sized material and colloidal material therefore overlap but are not identical.

The small size of colloids gives them a high surface-to-volume ratio. When a large particle is divided into many smaller particles, the total exposed surface increases greatly while the mass remains constant. More exposed surface means more sites for adsorption, ion exchange and chemical reaction.

Colloids may remain suspended in water for long periods because their settling is slow. Electrical charge may also cause particles with similar charges to repel one another. Suspension stability is controlled by gravitational settling, Brownian movement, electrostatic repulsion and attractive forces.

Important properties of soil colloids include high specific surface area, electrical charge, adsorption capacity, ion exchange capacity, hydration, swelling, shrinkage and aggregation. These properties are linked. Surface charge attracts ions; ions attract water; water changes interlayer spacing; and changes in spacing affect swelling and mechanical behaviour.

Most soil colloids have a net negative charge, but some surfaces become positive under acidic conditions. A negative surface attracts cations, whereas a positive surface attracts anions. The balance between these sites determines the chemical behaviour of a soil. (Iowa State University Digital Press, 2025)

2. Types and Mineral Structure

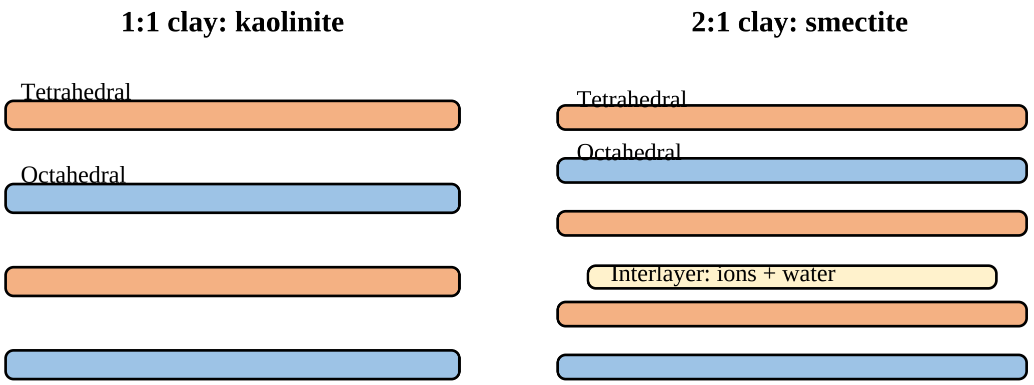

Figure 1. *Basic layer arrangement of 1:1 and 2:1 clay minerals.* Note. Original schematic based on Iowa State University Digital Press (2025).

Crystalline silicate clays are layer silicate minerals, also called phyllosilicates. Their structures are built from tetrahedral and octahedral sheets. A tetrahedral sheet contains silicon or aluminium surrounded by oxygen. An octahedral sheet contains aluminium, magnesium or iron surrounded by oxygen and hydroxyl groups.

Kaolinite is a 1:1 clay mineral because one tetrahedral sheet is joined to one octahedral sheet. Its layers are held together by strong hydrogen bonding. Water and ions do not easily enter between the layers, so kaolinite has little interlayer expansion and relatively low internal surface area.

Smectite is a 2:1 clay mineral containing one octahedral sheet between two tetrahedral sheets. Weak interlayer bonding allows water and exchangeable cations to enter the interlayer. Smectite therefore has high internal surface area, high water retention, high CEC and strong swelling and shrinkage behaviour.

Illite is also a 2:1 mineral, but potassium ions hold its layers more strongly than in smectite. Vermiculite has relatively high layer charge and can retain some interlayer cations strongly. These differences in structure produce differences in nutrient retention and physical behaviour.

Poorly crystalline minerals include allophane and imogolite. They are short-range-order minerals commonly found in volcanic soils. Their surfaces contain many hydroxyl groups and can develop strong pH-dependent charge. Iron and aluminium oxides also contain hydroxyl groups and are important for phosphate and other anion adsorption.

Organic colloids consist mainly of humus. Humus contains carboxyl, phenolic, hydroxyl and other functional groups. It contributes greatly to CEC, water retention, metal complexation and aggregate stability. (Iowa State University Digital Press, 2025)

3. Specific Surface Area and Surface Energy

Specific surface area is the total surface area of a material per unit mass, commonly expressed in square metres per gram. It increases as particle size decreases. This is the main physical reason why clay and humus are more chemically active than sand and coarse silt.

Colloids may have external and internal surfaces. External surfaces occur on the outside of particles and at crystal edges. Internal surfaces occur between layers that can separate or expand. Smectite has extensive internal surfaces because its interlayers can open, while kaolinite has mainly external surfaces because its layers are tightly bonded.

A large specific surface area provides more sites for adsorption of ions, water and organic molecules. It also provides locations for microorganisms to attach and for reactions to occur. A soil with more colloidal surface generally retains more water and nutrients, although strong retention may sometimes reduce immediate availability.

Surface energy is associated with atoms and ions at a particle boundary. Surface atoms do not have the same neighbours as atoms inside the crystal, creating an energetic imbalance. Adsorption of water, ions or organic molecules can reduce surface energy. This explains why colloids interact strongly with the surrounding solution.

Surface area alone does not determine adsorption. Two minerals may have similar surface areas but different charge densities, functional groups and crystal structures. Therefore, surface area, charge, mineralogy and solution composition must be considered together.

4. Origin of Surface Charge

Surface charge is an imbalance of positive and negative charges associated with a colloid. It controls which ions are attracted to the surface and influences attraction or repulsion between particles. Soil colloids may possess permanent charge, variable charge or both.

Permanent charge is mainly produced by isomorphous substitution inside crystalline silicate minerals. During crystal formation, an ion may be replaced by another ion of similar size but lower valence. For example, Al^{3+} may replace Si^{4+} in a tetrahedral sheet, or Mg^{2+} may replace Al^{3+} in an octahedral sheet. The substitution creates a negative structural charge that is comparatively independent of pH.

Variable charge develops at broken edges, mineral hydroxyl groups, iron and aluminium oxides, allophane and organic functional groups. These groups react with hydrogen ions. Simplified reactions are Mineral–OH + H^+ ■ Mineral–OH_2^+ and Mineral–OH ■ Mineral–O^- + H^+. The first reaction produces a positive site and the second produces a negative site.

At low pH, protonation commonly increases positive charge or reduces negative charge. At higher pH, deprotonation generally increases negative charge. Organic matter behaves similarly: R–COOH ■ R–COO^- + H^+. The point of zero charge is the pH at which positive and negative charges of a variable surface are balanced. (Virginia Department of Conservation and Recreation, n.d.)

Charge is also affected by organic and oxide coatings. Organic matter may block mineral sites or add new functional groups. Iron coatings may cement particles and provide new adsorption sites. Thus, the total charge of a soil is the combined result of mineral structure, surface reactions and organic matter.

5. Electrical Double Layer and Zeta Potential

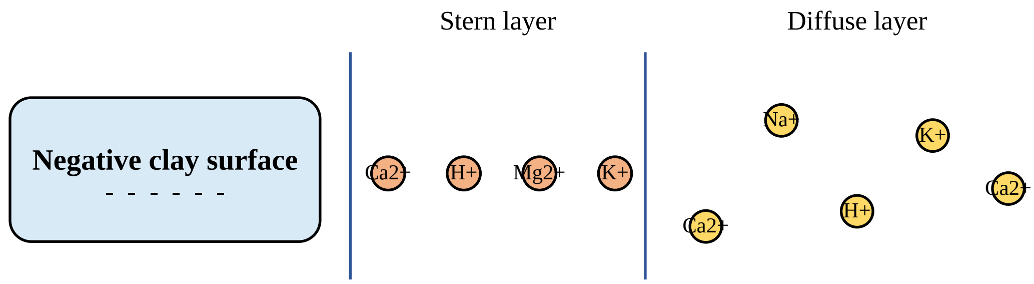

Figure 2. *Electrical double layer around a negatively charged clay surface.* Note. Original schematic based on the double-layer concept described by Wikipedia contributors (n.d.).

When a charged colloid is placed in water, ions arrange themselves around its surface. The surface charge and the compensating charge in solution form an electrical double layer. For a negatively charged clay, cations are concentrated close to the surface and anions are relatively excluded.

The inner part of the double layer is often called the Stern layer. It contains ions held relatively close to the surface. Outside it is the diffuse layer, where ions are more mobile and their concentration gradually approaches the concentration of the bulk soil solution.

The electrical potential is highest near the charged surface and decreases with distance. Zeta potential is the potential at the slipping plane between liquid moving with the particle and liquid outside that moving layer. A high absolute zeta potential generally indicates strong repulsion and stable dispersion; a lower absolute value allows particles to approach and aggregate. (Wikipedia contributors, n.d.)

Electrolyte concentration affects double-layer thickness. Increasing salt concentration compresses the diffuse layer and reduces repulsion. Dilute solution allows the layer to expand and can increase dispersion. Ion valence is also important: Ca^{2+} is more effective than Na^{+} in neutralising negative charge, while Al^{3+} may be even more effective.

The electrical double layer connects chemical conditions to soil physical behaviour. Changes in pH, salt concentration or exchangeable cations can change aggregation, pore continuity, hydraulic conductivity and erosion risk.

6. Adsorption, Desorption and Ion Exchange

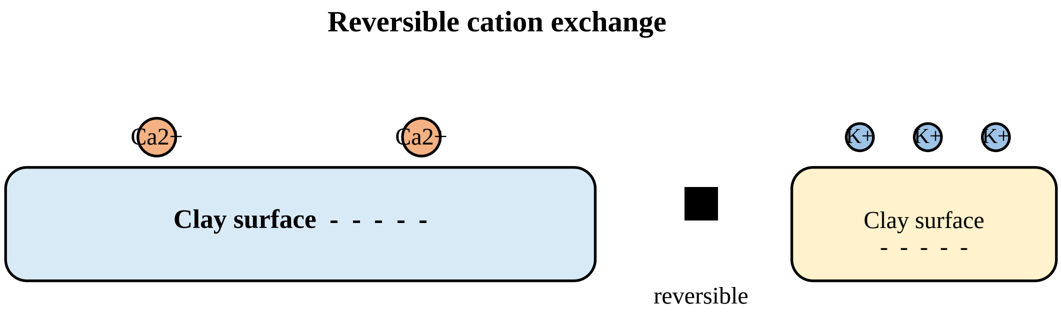

Figure 3. *Illustration of reversible cation exchange.* Note. Original schematic based on Iowa State University Digital Press (2025).

Adsorption is the accumulation of ions, molecules or water at a solid surface. It differs from absorption, in which a substance enters the interior of another material. Soil colloids adsorb substances through electrostatic attraction, chemical bonding, van der Waals forces, hydrogen bonding and ligand exchange. (Iowa State University Digital Press, 2025)

Electrostatic adsorption is caused by attraction between opposite charges. A negative clay surface attracts cations, while a positive oxide surface attracts anions. Specific adsorption is stronger and more selective because an ion forms a direct association with a surface functional group or replaces a surface hydroxyl.

Phosphate is a major example of specific adsorption. It forms strong associations with iron and aluminium oxides, especially in acidic and highly weathered soils. This may reduce phosphate leaching but can also reduce the amount immediately available to plants.

Desorption is the release of an adsorbed substance into the soil solution. Adsorption and desorption are dynamic. A nutrient may be adsorbed when solution concentration is high and desorbed when roots remove it. This dynamic reserve is important for plant nutrition.

Ion exchange is the reversible replacement of an adsorbed ion by another ion of the same charge type. A representative reaction is Clay–Ca2+ + 2K+ ■ Clay–2K+ + Ca2+. The reaction is electrically balanced because one calcium ion carries the same charge equivalent as two potassium ions.

7. Cation Exchange Capacity and Anion Exchange Capacity

Cation exchange capacity, or CEC, is the total quantity of exchangeable positive charge that a soil or colloid can retain. It is commonly expressed as cmolc kg–1. CEC is generally higher in soils containing smectite, vermiculite and humus than in soils dominated by kaolinite or crystalline iron and aluminium oxides. (Moorberg & Crouse, 2021)

CEC is important for nutrient management. Calcium, magnesium, potassium and ammonium can be retained on exchange sites rather than being immediately lost by leaching. The soil solution can be replenished when roots absorb ions or when solution concentration changes.

High CEC improves nutrient retention and buffering, but it does not automatically mean high fertility. Nutrients may be held too strongly, pH may be unsuitable, moisture may be inadequate, or one ion may interfere with another. A soil test should therefore be interpreted together with pH, base saturation, organic matter and mineralogy.

Anion exchange capacity, or AEC, is the quantity of exchangeable negative charge that a soil can retain. It becomes important where positive surface charge occurs, especially in acidic soils containing iron oxides, aluminium oxides, allophane and protonated clay edges. Phosphate, sulphate, nitrate and chloride may be retained on these sites.

CEC of variable-charge surfaces often increases with pH, while AEC often decreases with pH because deprotonation changes positive sites into negative sites. This opposite response is important in oxide-rich and volcanic soils.

8. Flocculation and Dispersion

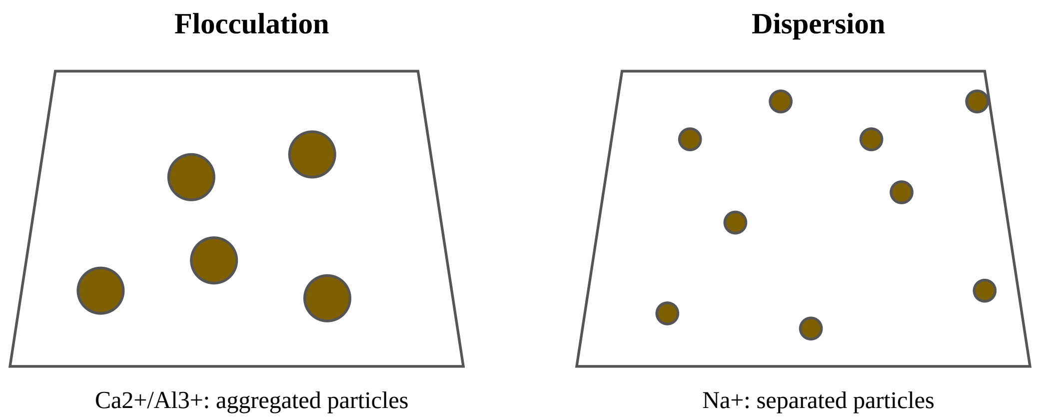

Figure 4. *Flocculation and dispersion of colloidal clay particles.* Note. Original schematic based on the soil colloid principles described by Moorberg and Crouse (2021).

Flocculation is the joining of colloidal particles into larger groups or flocs. Dispersion is the separation of particles from aggregates into individual units. The final condition depends on attractive forces, electrostatic repulsion, electrolyte concentration, exchangeable cations and surface chemistry.

Van der Waals forces attract particles. Electrostatic repulsion occurs when similarly charged diffuse layers overlap as particles approach. If attraction is stronger than repulsion, particles flocculate. If repulsion is stronger, particles remain dispersed. These ideas are commonly discussed through the principles of DLVO theory.

Calcium promotes flocculation because it is divalent and compresses the electrical double layer. Sodium is monovalent and usually produces less compression. A sodium-dominated clay therefore has a greater risk of dispersion, especially when soluble salts are leached away.

Flocculated soil contains larger aggregates and connected pores. It generally has better infiltration, aeration, root penetration and resistance to erosion. Dispersed clay can move into pores, clog them and form a crust at the soil surface.

Sodic soils demonstrate the connection between chemistry and structure. When sodium occupies many exchange sites and the soil solution becomes dilute, the double layer expands and dispersion becomes severe. Organic matter, roots and microbial products can help stabilise aggregates.

9. Swelling, Shrinkage and Mechanical Behaviour

Swelling occurs when water and hydrated ions enter the spaces between clay layers. It is strongest in expanding 2:1 minerals such as smectite because weak interlayer bonding allows layers to separate. Vermiculite can expand moderately, while kaolinite has strong layer bonding and little interlayer expansion.

Shrinkage occurs when water leaves the interlayer during drying. Clay layers move closer together and soil volume decreases. Repeated wetting and drying can produce cracks, changes in pore size, hard clods and movement beneath roads and foundations.

Swelling affects soil water relations. Some water is held on external surfaces, some around exchangeable cations and some between clay layers. Water held very strongly may be unavailable to plants. Total water content and plant-available water are therefore not identical.

Clay content, clay mineralogy, organic matter and exchangeable cations influence plasticity. Wet smectitic soil may become sticky and highly plastic. When dry, it may become hard and difficult to break. Kaolinitic and oxide-rich soils commonly have lower shrink–swell potential.

Colloids contribute to cohesion and adhesion. Cohesion is attraction between similar particles, while adhesion is attraction between different materials such as soil and water. These forces influence tillage, aggregate formation, crusting and wet-soil strength.

10. Factors Affecting Surface Behaviour

Mineralogy is a major control. Smectite has expandable interlayers and high internal surface area. Vermiculite may strongly retain some ions. Kaolinite has low expansion. Oxides and allophane have strong variable charge. Humus provides many pH-dependent functional groups.

Soil pH controls protonation and deprotonation of mineral and organic functional groups. Increasing pH commonly increases negative charge on humus and variable-charge mineral surfaces. Decreasing pH may increase positive charge on oxide surfaces and increase anion adsorption.

Electrolyte concentration controls diffuse-layer thickness. High dissolved salt concentration compresses the layer and encourages flocculation. Low concentration allows the layer to expand and may encourage dispersion. The effect is particularly important in irrigated and sodic soils.

Ion valence affects exchange selectivity and particle interaction. Calcium and magnesium generally stabilise clay more effectively than sodium. Hydrogen and aluminium influence acidity and may occupy exchange sites in acid soils. Sodium can contribute to aggregate breakdown when its proportion becomes excessive.

Organic matter increases surface area and adds functional groups. It can increase CEC, retain water, bind metals and improve aggregate stability. Moisture affects hydration, swelling, diffusion and reaction rates. Redox conditions can alter iron minerals, phosphate retention and trace-metal mobility.

11. Agricultural and Environmental Importance

Colloid surfaces retain calcium, magnesium, potassium and ammonium and release them through exchange. This reduces nutrient loss by leaching and creates a reserve that can replenish the soil solution. CEC is consequently an important part of fertiliser and nutrient management.

Clay and humus retain hydrogen and aluminium ions and resist sudden changes in soil reaction. Soils with more reactive colloids usually have greater buffering capacity, meaning that more lime or acid is required to produce a given pH change.

Phosphate may be strongly adsorbed by iron and aluminium oxides in acidic, highly weathered soils. This can reduce leaching but may also reduce immediate plant availability. Efficient phosphorus management requires attention to pH, oxide content, application rate and placement.

Colloids retain water and help form aggregates. Stable aggregates create pores for infiltration, air exchange and root growth. Dispersion blocks pores and may cause surface sealing, runoff and poor water entry. Organic matter and suitable exchangeable calcium help maintain stability.

Colloids can adsorb heavy metals, pesticides, phosphate and organic pollutants. Adsorption may slow movement to groundwater, but dispersed colloids can transport adsorbed materials in runoff. Expanding clays can damage roads, channels and foundations through repeated swelling and shrinkage.

12. Methods and Soil Management

CEC analysis measures the total exchangeable positive charge retained by a soil. A known index cation is used to saturate exchange sites and the amount displaced is measured. AEC analysis measures anion retention on positive sites and is useful for oxide-rich or volcanic soils.

Specific surface area can be measured using gas adsorption or other physical methods. Zeta-potential measurements evaluate dispersion stability. Potentiometric titration measures charge changes with pH and helps estimate the point of zero charge. X-ray diffraction identifies clay minerals and interlayer spacing.

Adsorption isotherms relate solution concentration to the amount retained by a solid. They are used to study phosphate, metals, pesticides and other compounds. Particle-size analysis, aggregate stability tests, water-retention curves and soil strength measurements connect surface chemistry with physical behaviour.

Soil management should maintain organic matter through plant residues, compost, cover crops and reduced disturbance. These practices improve aggregation, water retention and exchange capacity. Acid soils should be limed according to soil-test recommendations, because buffering capacity and

CEC affect the required rate.

Sodic and saline soils require careful irrigation and drainage management. Irrigation water should be tested for electrical conductivity and sodium hazard. Where appropriate, calcium amendments and leaching may replace exchangeable sodium. Fertiliser should be applied according to CEC, pH, mineralogy and crop demand. (Moorberg & Crouse, 2021)

Conclusion and APA 7 Source List

Soil colloids are the active surface fraction of soil. Their small size provides a large specific surface area, and their charge allows strong interactions with water, ions, organic compounds and microorganisms. Permanent charge develops mainly from isomorphous substitution, while variable charge develops from broken edges, hydroxyl groups, oxide surfaces and organic functional groups.

The electrical double layer explains the arrangement of ions around a charged colloid and the repulsion or attraction between particles. Adsorption retains ions and molecules at surfaces, while ion exchange stores and releases nutrient cations. CEC measures exchangeable positive charge, and positive variable-charge surfaces contribute to anion retention.

Flocculation produces larger aggregates and generally improves infiltration, aeration and resistance to erosion. Dispersion separates clay particles and may clog pores. Expanding minerals such as smectite absorb water between layers, causing swelling when wet and shrinkage when dry. These processes connect soil chemistry with soil physics, agriculture, environmental quality and engineering.

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Figure note. Figures 1–4 are original explanatory diagrams prepared for this assignment. The concepts are based on the phyllosilicate, colloid, sorption, ion-exchange and electrical-double-layer materials listed above. If additional external photographs are inserted, add creator, year, title, website and URL in an APA 7 figure note.